R Assignment 3-INFS 343

Part I Integer programming:

Consider the following IP problem.

$$Minimize \quad z = 9x_1 + 6x_2,$$

subject to:

$$\begin{aligned} &Constraint\ 1: && 5x_1 + 3x_2 \geq 30, \\ &Constraint\ 2: && 2x_1 + 5x_2 \geq 33, \\ &Constraint\ 3: && x_1, x_2 \geq 0, \\ &Constraint\ 4: && x_1, x_2 = \text{non-negative integer}, \end{aligned}$$

where x_1 and x_2 represent the decision variables. Solve this IP problem and answer the following questions.

a. **What are the values of X1 and X2 at the optimal solution?**
   X1 = 300
   X2 = 100

b. **What is the minimum value of z?**
   Z = 3950

c. **Please provide a screenshot of the R codes and answer the questions. (Note: use the comments function to put down your name and your Loyola campus ID in the R code)**

```
> library(lpSolve)
> # Marcelo Andrada Batista
> # 00001665216
>
> library(lpSolve)
>
> # Objective function (example)
> objective <- c(9, 12.5)
>
> # Constraints
> const.mat <- matrix(c(
+     1.5, 2,
+     2, 3,
+     8.5, 12.5
+ ), nrow = 3, byrow = TRUE)
>
> const.dir <- c("<=", "<=", "<=")
>
> const.rhs <- c(650, 900, 4000)
>
> # Solve
> result <- lp("max", objective, const.mat, const.dir, const.rhs)
>
> # Answers
> result$solution
[1] 300 100
> result$objval
[1] 3950
> |
```

d. If the objective function coefficients change to z = 8X1 + 7X2, what are the values of X1 and X2 at the optimal solution?

X1 = 433

X2 = 0

e. What is the minimum value of z based on the last step?

Z = 3464

f. Please provide a screenshot of the R codes and answer the questions. (Note: use the comments function to put down your name and your Loyola campus ID in the R code)

```
> # Marcelo Andrada Batista
> # 00001665216
>
> library(lpSolve)
>
> objective <- c(8, 7)
>
> const.mat <- matrix(c(
+     1.5, 2,
+     2, 3,
+     8.5, 12.5
+ ), nrow = 3, byrow = TRUE)
>
> const.dir <- c("<=", "<=", "<=")
> const.rhs <- c(650, 900, 4000)
>
> result <- lp("max", objective, const.mat, const.dir, const.rhs, all.int = TRUE)
>
> result$solution
[1] 433   0
> result$objval
[1] 3464
```

g. **Using the original objective function coefficients, if the second constraint changes to 2X1 + 4X2 >= 30, what are the values of X1 and X2 at the optimal solution?**

X1 = 72

X2 = 271

h. **What is the minimum value of z?**

Z = 4035.5

i. **Please provide a screenshot of the R codes and answer the questions. (Note: use the comments function to put down your name and your Loyola campus ID in the R code)**

```
> # Marcelo Andrada Batista
> # 00001665216
>
> library(lpSolve)
>
> # original objective function
> objective <- c(9, 12.5)
>
> # constraints (with changed second constraint)
> const.mat <- matrix(c(
+     1.5, 2,
+     2, 4,
+     8.5, 12.5
+ ), nrow = 3, byrow = TRUE)
>
> const.dir <- c("<=", ">=", "<=")
> const.rhs <- c(650, 30, 4000)
>
> # solve as integer programming
> result <- lp("max", objective, const.mat, const.dir, const.rhs, all.int = TRUE)
>
> # output
> result$solution
[1]  72 271
> result$objval
[1] 4035.5
>
```

j. **Did the optimal solution change? Why or why not?**

Yes, the optimal solution changed. This happened because we changed a constraint, which affects the feasible region and allows different solutions.

k. **What does this tell you about the sensitivity of the model to cost changes?**

It shows that the model is sensitive. Small changes in constraints or costs can change the optimal solution and the value of z.

Part II Workforce scheduling:

Workers at a local shipping port work on a five-day schedule followed by two days off. During the past few weeks, the shipping port has received a much higher volume of shipments, and the port manager wants to reexamine the staffing schedule. The accompanying table shows the number of workers that will be needed on each day.

Days	Workers needed
Monday	43
Tuesday	38
Wednesday	37
Thursday	30
Friday	31
Saturday	27
Sunday	25

a. **Write objective function (What are we minimizing?)**
   Minimize:
   x1 + x2 + x3 + x4 + x5 + x6 + x7

b. **Write constraints**

   x1 + x4 + x5 + x6 + x7 >= 43
   x1 + x2 + x5 + x6 + x7 >= 38
   x1 + x2 + x3 + x6 + x7 >= 37
   x1 + x2 + x3 + x4 + x7 >= 30
   x1 + x2 + x3 + x4 + x5 >= 31
   x2 + x3 + x4 + x5 + x6 >= 27
   x3 + x4 + x5 + x6 + x7 >= 25

   x1, x2, x3, x4, x5, x6, x7 >= 0

c. Formulate and solve the shipping port scheduling problem. Please provide the screenshots of the R codes. (Note: use the comments function to put down your name and your Loyola campus ID in the R code)

```
> # Marcelo Andrada Batista
> # 00001665216
>
> library(lpSolve)
>
> objective <- c(1,1,1,1,1,1,1)
>
> const.mat <- matrix(c(
+     1,0,0,1,1,1,1,
+     1,1,0,0,1,1,1,
+     1,1,1,0,0,1,1,
+     1,1,1,1,0,0,1,
+     1,1,1,1,1,0,0,
+     0,1,1,1,1,1,0,
+     0,0,1,1,1,1,1
+ ), nrow = 7, byrow = TRUE)
>
> const.dir <- c(">=", ">=", ">=", ">=", ">=", ">=", ">=")
>
> const.rhs <- c(43,38,37,30,31,27,25)
>
> result <- lp("min", objective, const.mat, const.dir, const.rhs, all.int = TRUE)
>
> result$solution
[1] 19  3  1  6  2 15  1
> result$objval
[1] 47
```

d. What is the minimum number of workers that the shipping port needs?

47 workers

e. Do more workers start on certain days? Which ones?

Yes. More workers start on Monday and Saturday.
Monday = 19 and Saturday = 15

While the other days start with way less.
Tuesday = 3 Wednesday = 1 Thursday = 6 Friday = 2 Sunday = 1

f. Why do you think more workers start on those days?

Because those days help cover the days with the highest number of workers needed.

g. How many workers need to start their five-day schedule on Saturday or Sunday?

16 workers

h. Give one real-world factor not included in this model. (hints: workers are human).

A real-world factor not included is that workers can get sick or miss work.